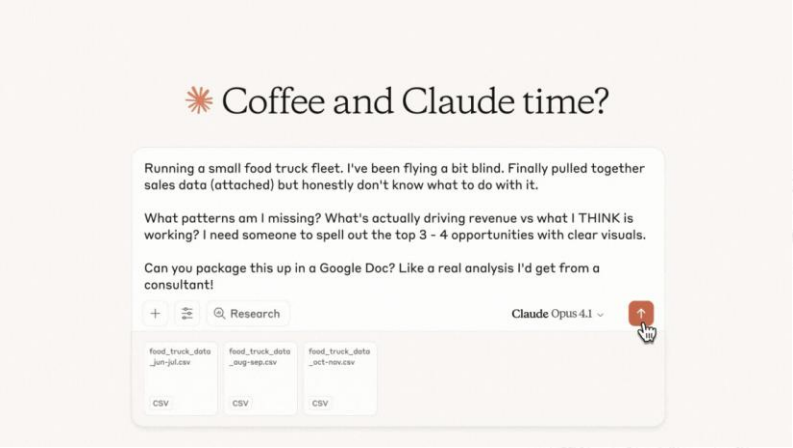
NIELIT Ropar & SCERT Punjab

Hands-On Demonstration

Enhancing Digital Productivity

using Advanced Word & Excel



Day 1 · Session 2 – Hands-On Demo

Teachers Training in ICT & Digital Initiatives

Claude AI is an advanced AI assistant developed by Anthropic. It is a large language model system designed for tasks like writing, coding, data analysis, and document generation

Hands-On

Word Demo 1

Styles & Templates

Word Demo 2

Mail Merge & Tables

Excel Demo 1

Formulas & Functions

Excel Demo 2

PivotTable & Charts

AI Demo

Copilot & Flash Fill

Q&A / Recap

Doubt clearing

WORD DEMO 1 — Interface & Setup

Before you begin: Open MS Word → Blank Document

Word Ribbon — Key Tabs to Know

Home	Styles, fonts, clipboard, paragraph formatting
Insert	Tables, pictures, headers/footers, page breaks
Design	Document themes, watermarks, page borders
Layout	Margins, orientation, columns, spacing
References	Table of Contents, footnotes, citations
Mailings	Mail Merge wizard, labels, envelopes
Review	Spell check, track changes, comments
View	Navigation Pane, Zoom, side-by-side view

Quick Setup Checklist

- ✓ Open Word → File → Options → check AutoSave is ON
- ✓ View tab → check ✓ Ruler and ✓ Navigation Pane
- ✓ Set margins: Layout → Margins → Normal (1 inch all sides)
- ✓ Set font default: Home → Font → Calibri 12pt
- ✓ Turn on Show/Hide ¶ (Ctrl+Shift+8) to see formatting marks
- ✓ File → Save As → create folder 'NIELIT_Training' on Desktop
- ✓ Quick Access Toolbar: add Save, Undo, Redo, Print Preview
- ✓ Status bar: right-click → enable Word Count & Page Number

WORD DEMO 1 — Styles, Heading & TOC

Live Task: Build a formatted School Annual Report skeleton

01 Type the Document Title

TYPE *Type: "GOVERNMENT SCHOOL — ANNUAL REPORT 2025-26"*

CLICK Home tab → Styles group → click Title style

SEE *Text becomes large, formatted title automatically*

02 Add Section Headings

TYPE *Press Enter → type "1. School Overview"*

CLICK Styles → Heading 1 | Add sub-heading → Heading 2

SEE *Hierarchical structure created — Navigation Pane updates live*

03 Insert Auto Table of Contents

TYPE *Press Ctrl+Home to go to top → Enter a blank line*

CLICK References tab → Table of Contents → Automatic Table 1

SEE *TOC generated instantly from all Heading styles!*

04 Update TOC after editing

TYPE *Add new heading "2. Academic Performance" anywhere*

CLICK Click inside TOC → Update Table → Update entire table

SEE *TOC refreshes automatically — no manual editing needed*

WORD DEMO 2 — Tables & Mail Merge

Live Task: Create a timetable table + personalised parent letters

Part A — Advanced Tables

Step 1 Type heading "CLASS TIMETABLE" → Heading 1

Step 2 Insert → Table → 6 columns × 7 rows (Mon–Sat + header)

Step 3 Select top row → Table Design → Header Row on, shade dark blue

Step 4 Type "Period 1–5" in columns, "Monday–Saturday" in rows

Step 5 Select entire table → Layout → AutoFit → AutoFit Window

Step 6 Merge top row cells: select → Layout → Merge Cells for title span

Step 7 Apply alternating row colour: Table Design → Banded Rows

Part B — Mail Merge for Parents

Step 1 Open Excel — create sheet: Name | Class | Marks | Grade

Step 2 Save as 'ParentData.xlsx' in NIELIT_Training folder

Step 3 Back in Word → Mailings → Start Mail Merge → Letters

Step 4 Select Recipients → Use Existing List → browse ParentData.xlsx

Step 5 Type letter. Click Insert Merge Field → select "Name" "Class"

Step 6 Insert field: "Dear <<Name>>, your ward in <<Class>> scored <<Marks>>"

Step 7 Preview Results → click arrows to see each parent's letter

Step 8 Finish & Merge → Edit Individual Documents → All

EXCEL DEMO 1 — Formula Masterclass

Live Task: Build a Student Marks Sheet — step by step from scratch

Demo Spreadsheet Structure

Name	Eng	Math	Sci	Soc	Total	Avg	Grade
Rahul K.	78	85	72	80	=SUM	=AVG	=IFS
Priya S.	92	88	95	91	=SUM	=AVG	=IFS
Amit V.	65	70	68	72	=SUM	=AVG	=IFS
Sneha R.	88	92	85	89	=SUM	=AVG	=IFS
Vijay M.	55	60	58	62	=SUM	=AVG	=IFS

Yellow cells = formulas to enter during demo

RANK formula in col I shows class rank

Type These Formulas — Step by Step

F2	=SUM(B2:E2)	Total marks (all 4 subjects)
G2	=AVERAGE(B2:E2)	Average — rounds to 2 decimal
H2	=IFS(G2>=90,"A+",G2>=80,"A",G2>=70,"B+",G2>=60,"B",G2>=50,"C",TRUE,"F")	Grade using IFS
I2	=RANK(G2,$G$2:$G$21,0)	Class rank (0 = descending)
J2	=IF(G2>=50,"PASS","FAIL")	Pass/Fail status
F22	=MAX(F2:F21)	Highest total in class
F23	=MIN(F2:F21)	Lowest total in class
F24	=COUNTIF(H2:H21,"A+")	Count of A+ students

EXCEL DEMO 2 — XLOOKUP & Data Validation

Live Task: Student search system + dropdown input validation

Part A — XLOOKUP Search System

1. **Create Search Sheet**
   New sheet tab → rename "Search". In B2: type label "Enter Student Name:"
2. **Type Student Name in C2**
   C2 is the search cell. Type any student name to trigger the lookup dynamically.
3. **XLOOKUP for English marks**
   In C4 type: =XLOOKUP(C2,Sheet1!A2:A21,Sheet1!B2:B21,"Not Found")
4. **Repeat for each subject**
   Copy C4 formula → paste in C5:C8 → adjust column reference B→C→D→E→F
5. **Fetch Grade & Total**
   C9: =XLOOKUP(C2,Sheet1!A:A,Sheet1!H:H,"Not Found") — fetches grade
6. **Test with different names**
   Change name in C2 — all results update INSTANTLY. Try a wrong name — see 'Not Found'

Part B — Data Validation

- **Dropdown List**
  Select Grade column → Data → Data Validation → Allow: List → enter A+,A,B+,B,C,F
- **Whole Number Range**
  Select Marks columns → Validation → Whole number → between 0 and 100. Error alert: 'Invalid marks!'
- **Input Message**
  Data Validation → Input Message tab → Title: 'Marks Entry' → Message: 'Enter value 0–100'
- **Error Alert**
  Error Alert tab → Style: Stop → Title: 'Invalid!' → Message: 'Marks must be between 0 and 100'
- **Circle Invalid Data**
  Data tab → Data Validation → Circle Invalid Data — red circles appear on wrong entries!
- **Conditional Formatting**
  Select Avg column → Home → Cond. Format → Color Scale → Green-Yellow-Red gradient

EXCEL DEMO 3 — PivotTable & Charts

Live Task: Analyse marks data visually — PivotTable + Bar Chart + Sparklines

PivotTable — 7 Steps

1. Click any cell inside your marks table
2. Insert → PivotTable → New Worksheet → OK
3. Drag 'Grade' to Rows field area
4. Drag 'Total' to Values area → Sum of Total
5. Right-click Values → Value Field Settings → Average
6. Insert → Slicer → select 'Grade' for interactive filter
7. Click PivotTable → PivotChart → Clustered Bar → OK

Charts & Sparklines

Bar Chart — Subject-wise

Select subject headers + 5 student names → Insert → Bar Chart → 2D Clustered Bar. Title: 'Subject Performance'

Pie Chart — Grade Distribution

Select Grade & Count data from PivotTable → Insert → Pie Chart → Show Data Labels & Percentages

Line Sparklines per student

Select cell K2:K21 → Insert → Sparklines → Line → Data Range B2:E21. See mini trend per row!

Format Chart Professionally

Click chart → Chart Design → Style 8 (dark). Add data labels, title, legend. Change colors to match school theme.

Conditional Formatting — Heat Map

Select all marks B2:E21 → Home → Cond. Format → Color Scale → 3-Color scale (red-yellow-green)

AI DEMO — Flash Fill, Copilot & Analyze Data

Live AI demonstrations in both Word and Excel

⚡ Excel: Flash Fill (AI Pattern Detection)

- Column A has full names: "Rahul Kumar", "Priya Singh"...
- In B2, type just the first name: "Rahul" → press Enter
- In B3, type "Priya" → Excel senses pattern!
- Press Ctrl+E → Flash Fill completes entire column!
- Try Column C: type last names only → Ctrl+E again
- Try column D: type "R. Kumar" → Ctrl+E → initials format!

Excel: Analyze Data (AI Insights)

- Click any cell in your marks dataset
- Home tab → Click "Analyze Data" (right side of ribbon)
- AI task pane opens — shows auto-generated insights
- See: Top performers, subject averages, outliers
- Click any suggestion → inserts PivotTable automatically!
- Ask: "Which subject has lowest average?" → instant answer

Word: Editor + Copilot Demo

- Open the Annual Report document you made earlier
- Home → Editor → see grammar & style suggestions
- Accept suggestions — watch text improve in real time
- Select intro paragraph → Right-click → Copilot: Rewrite
- Copilot prompt: "Make this more formal and concise"
- Try: "Summarize this document in 5 bullet points"

Excel: Forecast Sheet (AI Prediction)

- Create 2-column data: Month | Avg Score (6 months)
- Example: Jan:68, Feb:71, Mar:74, Apr:72, May:76, Jun:79
- Select both columns of data
- Data tab → Forecast Sheet button
- Preview shows predicted next 3 months with bands!
- Click Create → new sheet with chart + predictions

⚠ Common Errors & How to Fix Them

Things that go wrong — and the exact fix during hands-on practice

#VALUE! — Excel

Fix: Wrong data type in formula. Check: no text cells in SUM/AVERAGE range. Use ISNUMBER() to debug.

#N/A — Excel

Fix: XLOOKUP didn't find value. Add 4th argument: 'Not Found'. Check spelling of search term.

Mail Merge fields blank — Word

Fix: Excel file not saved. Data source column names must match field names. Re-select recipient list.

Table of Contents wrong — Word

Fix: Right-click TOC → Update Field → Update entire table. Make sure no extra Heading styles applied.

#REF! — Excel

Fix: Cell reference deleted. Re-enter formula. Use absolute refs ($A$1) before deleting rows/columns.

TOC shows no headings — Word

Fix: You forgot to apply Heading styles. Select each heading → Home → Styles → Heading 1/2.

Flash Fill not working — Excel

Fix: Flash Fill may be off. File → Options → Advanced → check 'Automatically Flash Fill'. Or use Ctrl+E.

Chart not updating — Excel

Fix: Click chart → Chart Design → Refresh Data. Or right-click → Select Data → re-select range.

🏆 Hands-On Challenge — Try It Yourself!

Complete both challenges before the tea break at 11:15 AM

WORD CHALLENGE

"School Exam Circular with Mail Merge"

1. Create a new Word document
2. Title: 'EXAMINATION SCHEDULE — MARCH 2026' (Title style)
3. Add 3 sections with Heading 1: Exam Rules, Schedule, Instructions
4. Insert a 6×5 table: Date | Day | Subject | Time | Room
5. Add 5 exam entries with proper formatting
6. Generate auto Table of Contents at top
7. Header: School name | Footer: Page number + date
8. Use Editor AI to check and improve the language
9. Mail Merge: Create 5 parent records in Excel → merge letter
10. Export final document as PDF

EXCEL CHALLENGE

"Class Performance Dashboard"

1. Create sheet with 20 students, 5 subjects
2. Add: Total (SUM), Average (AVERAGE), Grade (IFS)
3. Add: Pass/Fail (IF), Rank (RANK), Highest (MAX)
4. XLOOKUP search box — search any student by name
5. Data Validation: Dropdown for Grade, 0-100 for marks
6. Conditional Format: Red<50, Yellow 50-75, Green>75
7. PivotTable: Average marks by Grade category
8. Bar chart comparing subject averages across class
9. Add Sparklines in last column for each student trend
10. Use Analyze Data AI to get 3 auto-generated insights

Session 2: Key Takeaways

Word Mastery

You can now build styled documents with auto TOC, professional tables, and personalised mail-merged letters — in under 10 minutes.

Excel Power

From SUM to XLOOKUP to PivotTables — your data can now be searched, summarised, and visualised without manual effort.

AI Acceleration

Flash Fill detects patterns, Analyze Data gives instant insights, Copilot rewrites and summarises — AI is your productivity co-pilot.

You Saved Time

Every shortcut, every formula, every AI feature you used today translates to hours saved each week in your teaching workflow.